

Innovation in management education: adding value with simulation-based learning

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Abstract

Management problems are dynamic and often novel; simulation-based learning contributes to learning ‘how-to-learn’, a skill that is conducive to life-long learning and capabilities for innovation on the job. The impact of simulations on university-level learning has been investigated most thoroughly in medical education. Simulations can also make an important contribution to the development of the new skills demanded of contemporary management graduates in knowledge-based economies. We review how interactive stochastic simulations integrate core findings from research on learning in order to improve the quality and kinds of learning that can be achieved. For example, using the example of decision cases run on a market simulator, we highlight the importance of immediate feedback and practice for skill acquisition. The market aggregates participants’ decisions and provides immediate feedback concerning the success of a strategy. Analyzing the consequences of their decisions in different situations (potential future outcomes), allows participants to adapt their strategy in real time. Multiple replications of the case simulation allow participants to practice finding a robust strategy for the types of risks and opportunities they face for a particular task.

Background

Business leaders, governments and academic organizations have stressed the economic and social importance of innovation. The AACSB Task Force Report on innovation (2010) articulated ways that managers and business schools can promote innovation. The statement of general educational principles from the global accreditation body for business schools (AACSB, 2018) calls for innovation in the *delivery* of teaching -- emphasizing engagement, innovation and impact.

In a complementary proposal, Rousseau and McCarthy (2007) argued for innovation in *curricular content*, promoting the importance of evidence-based practice. The systematic use of data and research to guide improvements in safety and efficiency in health care is well-known and has been achieved through ‘best practice’ guidelines based on rigorous reviews of the research literature, and by prospective data collection and analysis. Their work has generated interesting developments in curricular content (for example, developing skills such as critical appraisal and the use of systematic reviews).

Despite the appeal of these evidence-based approaches for educating students, they are best suited for teaching students to apply evidence and best practices to known and well-researched problems and tasks. They do not focus on preparing students to solve new problems or to apply their existing knowledge and skills creatively – in other words, to prepare students for *ongoing* learning and innovation. As *The Lisbon Strategy – Making Change Happen* (Commission of European Communities, 2002: 7) highlights, this capability is critical for the economy:

“Research, innovation and education are at the heart of the knowledge-based economy. Europe’s ability to produce, diffuse and use knowledge effectively relies heavily on its capacity to carry out quality research, to produce highly educated people and for its firms to be engaged in a continuing process of innovation.”

Given the importance of a continuing process of innovation for the economy, how can management education best instill in students the capability for on-going innovation in their work? One approach has been to develop programs that focus explicitly on teaching about how to manage innovation processes and activities. Many business success stories focus on the impact of a particular innovation to generate productivity gains and competitive advantage. But innovations are inherently unpredictable and likely to produce novel situations and problems. Curricula based on evidence-based content and management of innovation do not necessarily promote the capacity to respond to fundamentally new problems in innovative ways.

The next section summarizes some ways that simulation-based learning can build students’ capacity for innovation. We then make a case for *interactive* and *stochastic* simulations, using the example of a market simulator with decision cases custom designed to promote learning how-to-learn and self-directed learning when faced with complexity and material uncertainty. This is followed by a section summarizing some evidence for the benefits of simulation-based learning. Evidence of a different type, that is, the educational rationale that explains the mechanisms through which simulations benefit learning, based on core findings from research on learning, is summarized in the next section and subsections. Finally, we offer some concluding comments.

Building Students’ Capacity for Innovation

The capacity to build new expertise and rapidly adjust to opportunities and change has been described as ‘flexpertise’ (Van der Heijden, 2002) and ‘adaptive expertise’ (VanLehn & Chi, 2012). In short, students need to learn ‘how to learn’, and how to learn from experience. This requires students to develop their capacity to learn independently and to continue do so throughout their careers. To do so, they need to learn to monitor their own behavior, track progress and make plans

i.e., to independently direct and control their own learning (De Jong & van Joolingen, 1998). Teaching goals and methods should explicitly foster and enhance these capacities.

In many professional education programs, supervised experience in real-world settings (experiential learning) is used to help students learn and apply their knowledge and skills. Management education also has a tradition of using classroom-based strategies to approximate aspects of practical experience (e.g. case studies and business games). Many of the latter (for example, Capstone, Foundation, The Business Strategy Game and GLO-BUS) are competitive games that build on case study methods by requiring teams of students to manage a simulated firm. However, these methods provide limited opportunities for practice (Rousseau and McCarthy, 2007) or the associated adaptation of strategies required for innovation. These methods are also not well-suited to disciplines which require high levels of technical skills in which current practice is embedded in technology with material uncertainty and operational risks.

In sectors such as aviation and medicine, where the costs and risks of learning on the job are high (e.g. slow responses and decision errors can have catastrophic consequences), skills and experience are developed in controlled environments such as simulators. In social science applications, such as finance and economics, the risks and uncertainties associated with decisions are even more complex since they are also affected by model uncertainty, parameter uncertainty, and signal extraction issues for the varying signal-to-noise ratios associated with information required for decisions. For example, Fama and French (2010) use bootstrap simulations from long histories of fund returns in order to confidently identify skill versus luck associated with portfolio managers' performance. This research example highlights the value of using simulations to learn how to separate signal from noise, an important skill for many empirical finance decisions.

Early reports of computer-based simulations in management education described simple simulations developed for a single purpose. For example, Berg, Dickaut, Hughes, McCabe and Rayburn (1995) described a simulation of an asset-market to demonstrate how prices are influenced by accounting information. Berg *et al.* articulated several advantages of a computer-based simulation including: resemblance between the simulation and actual institutions and assets (realism); ease of administration and participation (simplicity); and the ability to engage students by capturing their attention and providing prompt feedback on their actions (engagement). As with much educational software, the development costs and learning curve discourage widespread uptake – particularly

when the simulation is limited to just one course topic. Perhaps this explains Clarke's observation that business simulation games are primarily used in upper year and capstone courses (2009).

More recent advances with technology have included labs with real-time data feeds, access to historical databases, and practice with the use of industry-standard tools (for example, Alexander *et al.*, 2001; Holowczak, 2005, 2007; Coe, 2007; Coe, Kilic & Isik, 2007; Siam, 2005). These allow students to practice using information and applications from industry. However, these resources were designed for professional users rather than as educational applications that have been custom-designed to achieve specific learning objectives. For example, students need to learn how to deal with uncertainty by practicing across a wide range of potential outcomes. Simulation cases can be designed to explicitly incorporate uncertainty about the future.

Interactive Stochastic Simulations

Interactive stochastic simulations designed specifically for management education are a significant advance. Simulations that require students to apply knowledge to achieve specific management objectives are especially useful for acquiring skills and the capacity for innovation when faced with complexity and uncertainty; and can be designed to complement classroom teaching of theory. As outlined in the next section, interactive stochastic simulations integrate core findings from research on learning. We argue that it is important to include both stochastic and interactive features in the simulation cases.

The RIT market simulator and the associated decision cases (Mak and McCurdy, 2019) can be used to practice dealing with both exogenous uncertainty and the endogenous uncertainty introduced by market participants' actions. During each replication of the simulation, relevant variables change stochastically (randomly) with certain probabilities, reflecting fundamental uncertainty about future outcomes. The market aggregates participants' decisions and provides immediate feedback concerning the success of a strategy. Analyzing the consequences of their decisions in different situations (potential future outcomes), allows participants to adapt their strategy in real time. Multiple replications of the case simulation allow participants to practice finding a robust strategy.

In addition to modeling uncertainty explicitly, it is also important that the simulations are interactive: students should not just watch a simulation with no opportunity to interact with, or potentially affect, the simulation itself. The RIT market simulator allows students' decisions to benefit from interactions with the market (and other participants) in several ways; allowing them to

learn from the actions of others and to experience potential behavioral effects on decision making in an uncertain environment. Real-time datalinks from the market to participants' decision-support models allow students to update their decisions in real time as situations change. Decisions are submitted to the market manually or automatically through an application program interface (API), potentially affecting market outcomes.

Interactive stochastic simulation cases can be used for a variety of educational purposes throughout the curriculum, similar to the way that flight simulators are used for training pilots across the continuum from basic flying skills through recurrent training of professional pilots. Simulations can prepare students for work in the real world; allowing them to practice deriving innovative solutions to problems. They extend students' learning beyond 'textbook' examples by providing extensive practice with ill-defined problems in uncertain, dynamic environments similar to those in real world business environments. In addition, by providing instantaneous feedback to guide adaptation of strategies, simulations promote self-directed learning essential for learning on the job, a skill friendly to innovation.

Replications of the simulation based on the range of potential scenarios allows students to develop and practice skills to identify and assess risk and opportunities associated with their assigned task. The dynamic and uncertain outcomes, paired with immediate feedback on students' decisions, provide a powerful demonstration of the limitations of relying on rote, rule-of-thumb, solutions versus applying and adapting decision-support models. Performance feedback tools can also be used to elicit discussion about the fundamental drivers of good results and allow instructors to provide incentives for participants to focus on the learning objectives of each case. By engaging students in the decision support modeling and improving their decisions after each replication, the RIT decision cases promote learning how-to-learn, a skill that is conducive to life-long learning and innovation.

Flight simulators are used to acquire skills for fast decision making and implementing automation for some tasks. Automated systems can be efficient but fragile which is why skills for monitoring and responding quickly to problems are so crucial. As noted above, the RIT package has built-in APIs that allow participants to practice writing simple scripts for implementing some decisions automatically. Simulations can be sequenced such that participants make the decisions manually first -- so that they fully understand the scope of potential outcomes -- before writing an algorithm and implementing strategies automatically. RIT simulation cases are also designed to sequence skill acquisition, beginning with cases that focus on mastering single skills (dealing with one risk) and

progressively adding risks across a set of cases on a particular topic until student can manage a capstone case in which several risks interact (cascading risks in the flight simulator analogy). The market simulation platform and associated decision cases provide a flexible and powerful learning environment that can be used across the curriculum.

Evidence for the Benefits of Simulation-Based Learning

The impact of simulations on university-level learning has been investigated most thoroughly in medical and health professional education; several systematic reviews have shown benefits of simulation-based learning in medical and health professional education. These benefits include: consistent gains in students' performance and confidence associated with the use of simulations for learning (Harder, 2010); improved skills acquisition in several domains (for example, procedural, cognitive, group skills), and, of even greater importance, transfer of skills learned through simulation methods to clinical practice (McGahie, Issenberg, Petrusa and Scalese, 2010). Consistent with these findings, a systematic review and meta-analysis provided by Cook et al. (2011) concludes that "in comparison with no intervention, technology-enhanced simulation training in health professions education is consistently associated with large effects for outcomes of knowledge, skills, and behaviors and moderate effects for patient-related outcomes". We note that effect sizes are rarely considered in research on teaching methods in higher education, but they are important when deciding on the costs and benefits of educational innovations: the large effect sizes reported for technology-enhanced simulation training suggest that the benefits are substantive.

Further, a systematic review of medical simulations by Issenberg et al. (2005) gives insight into the conditions under which simulations benefit learning. These conditions include: integration into curricula and capture of a variety of (clinical) conditions; clear learning goals and outcomes; a range of difficulty levels and adaptable to multiple learning strategies; repetitive practice and feedback; active participation in learning; and a safe environment for errors. Instructors should carefully evaluate whether simulation packages are designed to fulfil these conditions. (The description of the RIT market simulator platform and Decision Cases above illustrates how simulations can be used to fulfill these conditions in management education.)

It is difficult to find empirical and comparative research on the impact of simulations in the management education literature. Salas et al. (2009) discuss practical benefits associated with simulation-based training for management education, including: providing a more complex and realistic learning environment than other training strategies while still allowing reality to be

simplified and manageable; providing a (relatively) risk-free environment for learning and experimentation; leading to learning in reduced time; providing an ideal method for training infrequently engaged but critical skills; is more engaging than other training methods; and can impart complex applied competencies. The majority of studies have relied on students' perceptions and self-reported learning (Clarke, 2009) although some have used relatively sophisticated methods (e.g. Ranchhod, Gurau, Loukis and Tirvedi (2014) reported positive relationships between students' perceptions of their experience and self-reported development of conceptual understanding and skills with a marketing business simulation game).

As preliminary evidence of how students' experience is impacted by interactive stochastic simulations, we report some results from student surveys and a course evaluation for the first immersive course taught using the RIT Market Simulator and Decision Cases. A 5-question survey was distributed to students who had been enrolled in classes using a lab with simulation-based learning applications. They were asked whether they strongly agree, agree, neither agree nor disagree, disagree, or strongly disagree with each question. The most telling statistics are that 89% of students either strongly agree or agree that the integration of simulation-based learning applications in their course(s) provided a superior learning experience compared to the conventional course approach; and that 90% of students agreed or strongly agreed that the simulation-based exercises are valuable for future students. The results also showed that 90% of students strongly agree or agree that the interactive learning exercises increased their engagement and interest in the course material. In terms of student collaboration, 88% of students agreed that they their experience fostered a collaborative learning environment. When students were questioned about job relevance, 73% responded favorably to this question with 19% of students neither agreeing nor disagreeing. This is symptomatic of the fact that not all students intend to go into directly-related careers.

Course evaluations are consistent with this pattern of positive perceptions. For example, evaluations from the first MBA and Commerce courses that used simulation-based learning in an immersive way, resulted in average ratings that were significantly above program averages. Representative comments highlighted experience and learning outcomes that were consistent with factors we have identified as important above. For example:

"This was my favorite course in the program so far. I felt it finally integrated most theory so I understand it on a more practical level. The ... cases created a lot of learning opportunity – as to do well, you really had to understand the background. I would recommend this course to anyone."

"I enjoyed the course throughout, including all the cases. They were great as a step by step introduction to the topics that are important ..."

“Excellent use of experiential learning. Interactive cases very well integrated with learning objectives. Assignments provide good opportunity to think about and apply concepts ... at various levels”

Simulation-based learning has transformative potential: it facilitates development of capabilities that are difficult or impossible to develop using conventional educational strategies. These cannot be captured using simple comparisons between simulation-based learning and conventional teaching. The next section provides evidence of a different type, that is, the educational rationale that explains the mechanisms through which simulation-based learning benefits learning.

Educational Rationale for Simulation-Based Learning

We have argued that interactive simulations can make a strong contribution to contemporary management education and summarized how well-planned interactive simulations can advance the goal of educating management graduates to be innovative on the job. In the following subsections, we will describe how interactive simulations embed educational principles highlighted by the AACSB Standards (and other research on student learning) to strengthen and extend student learning in ways that contribute to the capacity for innovation.

Provides Motivation

Educational researchers have become interested in simulations of real-world tasks for learning precisely because the ability to use knowledge in the real world is so important in a knowledge-based economy. Learning to understand and apply difficult concepts and complex skills requires effort and can be very challenging, so methods that increase motivation to learn can be very helpful. Voogt and van den Akker (2002) comment that simulations can ‘...serve as a bridge between reality and the student’s mental model of reality’. This can help students see the real-world relevance of abstract theories and overcome their resistance to learning abstract material. Lombardi (2007) highlighted additional educational benefits of connecting learning with real world problems, including improvements in student engagement, increased motivation to persist with practice to achieve high levels of performance, integration of different kinds of learning and skills, improvements in communication and collaborative skills, and increased ability to recognize and solve novel problems.

Structures Learning

Learning how to apply theoretical knowledge to management tasks and scenarios can be challenging. Students must clarify their understanding, work out the connections between their

theoretical understanding and relevant aspects of the problem, and determine how these connections inform actions. Interactive simulations can provide several additional types of support to help students meet the cognitive demands of understanding and applying new knowledge. Simulations are designed to remove unnecessary detail so that students can focus attention on elements that are relevant to a specific learning objective. This is especially valuable because novice problem solvers have difficulty distinguishing between superficial details and the fundamental variables (Chi, Feltovich and Glaser, 1981). As noted above, simulations can also be designed to sequence skill acquisition. In addition, unlike deterministic or non-stochastic models, simulations can provide multiple replications drawn from the range of potential scenarios. This not only leads to a better appreciation of the implications of uncertainty but also allows students to practice deriving strategies that work well systematically, that is, across the whole range of potential outcomes.

Facilitates Deep as Opposed to Surface Learning

Students must have a strong knowledge base that they can apply to solve new problems (Chi, Glaser and Rees, 1982). A significant body of research (starting with Marton and Säljö, 1976) has demonstrated that university students hold varying conceptions about knowledge and how it is acquired, and that this influences their learning. Students with a *deep* approach to learning are interested in the meaning and interrelationships of the ideas about which they are learning. Subsequent research has shown that that by approaching learning in this way, deep learners tend to achieve better understanding of material, and to remember it better, than do students who report a *surface* approach to learning (emphasizing memorization without necessarily understanding the material).

Students' Approaches to Learning can be adapted in response to course design factors under the control of instructors. Instructional approaches and assessments that actively involve learners in *working with* the material (applying it or learning-by-doing) foster deep learning require students to clarify their understanding of ideas, to relate them to what they already know, and to use the new material in meaningful ways. Such methods direct students' attention and learning strategies to the deep cognitive processes required to understand and apply or adapt material creatively to novel problems – capacities which are fundamental for innovation.¹

Simulations that provide variations on simulation scenarios and parameters can reinforce and extend

¹ Note that 'active' learning is not necessarily conducive to deep learning. Students can be actively engaged in superficial tasks with little if any educational benefit!

learning beyond the confines of solving a limited number of textbook problems. This can enrich students understanding and ability to apply and adapt models by allowing them to test their initial understanding and to generate and try out alternative ideas to observe the consequences. Engaging students in actively generating and testing new and alternative ideas can foster the generative thinking and the self-regulatory processes required for life-long learning and innovation.

Clarifies Concepts that are Difficult to Learn

Simulations can be especially helpful for teaching concepts which are especially difficult for students to grasp. Meyer and Land (2005) noted that some central disciplinary concepts are intrinsically more difficult to learn than others (or may need to be built up over time and cannot be comprehended immediately); such difficult concepts can be identified through recurrent patterns of errors and misconceptions in student work. The characteristics that make these concepts difficult to learn include counterintuitive, complex or abstract ideas, and interactions between two or more variables (as in many management models). Wright and Hibbert (2015) point out that Threshold Concepts have received limited attention in management education and suggest that they can offer useful insights for curriculum design and teaching. Examples identified for introductory management education include the relevance of theory for management practice (Wright and Gilmore, 2012) and for specialized fields such as quantitative finance examples include: ‘change of measure’, Diamond and Smith (2011); information asymmetry, liquidity, time value of money, risk versus return, and arbitrage, Hoadley, Tickle, Wood and Kyng, (2015). Identification of such concepts enables instructors to focus students’ time and attention where the payoff -- in terms of improvements in student understanding of difficult but important concepts in management -- is likely to be greatest. Simulations provide an especially powerful way to direct and facilitate learning of these concepts. For example, using RIT simulation cases to learn how to manage liquidity risk provides a very effective way for students to grasp the concept of a ‘marketable limit order’ which is an essential strategy for minimizing price impact (a cost of illiquidity in the market).

Increases Opportunities for Practice

As Vermundt (2003) pointed out, students must be given early and frequent opportunities to connect theory with practical experience so that the two kinds of knowledge can be integrated into a coherent knowledge base. Performing a new task requires students to process a lot of information – they must keep the theory in mind and make connections between it and the simulation – i.e. new tasks carry a high intrinsic cognitive load (Sweller, 1994). Practice helps students become familiar

with the material and reduces the intrinsic cognitive load, thereby freeing up cognitive resources for deep learning. Increasing the relative proportion of cognitive resources allocated to deep learning can result in students learning more quickly and/or more effectively. Simulations can be used to ‘scaffold’ learning, moving from early guided practice to progressively independent practice of new skills.

Allows Immediate Feedback and Assessment

Feedback in simulations can be both a stimulus for, and part of, the process of learning. Interactive simulations can improve learning by providing instant and ongoing formative feedback for students. The findings from meta-analyses of research on feedback (for example, Hattie and Timperley, 2007) report that, in appropriate conditions, the gains associated with formative feedback are substantially larger than those associated with many other interventions and variables related to learning.

Of course, management education already incorporates extensive faculty and peer feedback. However, simulations provide a complementary source of rich feedback without adding to already heavy faculty workloads: feedback is intrinsic in the way that simulation data, models and systems respond to students’ actions.

Computer-based simulations can be embedded throughout courses to augment conventional teaching and assessment. It is easy to vary task parameters for assessment purposes, and to track and report students’ performance. The flexibility adds rigor, reliability and efficiency to assessments. These advantages can in turn be used to improve learning by increasing the frequency of assessments, which encourages students to spend more time ‘on task’ and to distribute it evenly across the term (Gibbs and Simpson, 2004), without adding to instructors’ workload. Immediate records of students’ simulation performance also allow instructors to monitor learning more closely and to respond quickly when students run into difficulty.

Encourages Learning How-to-Learn

Academic courses direct students’ learning but often do little to prepare students to continue to learn independently as professionals. Interactive simulations can help this important transition by providing an ideal environment for students to learn how to use feedback to drive further learning, i.e. ‘feedforward’. In a complex decision-making environment, immediate feedback concerning the outcome of decisions, and how they compare to other participants and to the designated benchmark,

provides critical information which students can monitor and use to improve their performance (that is, to develop the capability for self-regulated independent learning). Encouraging students to take responsibility for assessing their work and using this information to direct their future learning is important for several reasons. The most immediate reason is that the skills of self-assessment and independent, self-regulated learning are an important component of learning-how-to-learn and will contribute to on-the-job (lifetime) learning. It is especially important to develop these skills in management education because they are so rarely addressed elsewhere in higher education.

Allowing students access to interactive simulations can also encourage independent learning by allowing them to learn at their own pace and at their own level or interest. In their call for a culture of evidence-based management, Rousseau and McCarthy (2007) relate the need for management education to inculcate the use of evidence for self-assessment and for finding ways to improve performance. Immediate feedback from simulations is an intrinsic source of evidence that encourages students to assess and adapt skills and strategies.

CONCLUDING COMMENTS

This article responds to recent calls for evidence-based innovation in management teaching and for business schools to increase their students' capabilities for promoting and managing innovation. Students need to go beyond learning to understand material conceptually; they need to learn how to use their knowledge and skills in complex, real-time environments with uncertainty. Since the management problems associated with innovations are dynamic and often novel, students need to learn 'how to learn' and to make high level evaluation of their own and others' performance. This requires learning regulatory processes to direct and control one's own learning (for example, monitoring one's own behavior, tracking progress, and advance planning).

The AACSB Task Report (2010) referred to the need for business education to adopt pedagogical methods better suited to help students develop the skills to use what they have learned when they enter the workplace. Similarly, Rousseau and McCarthy (2007) advocate the necessity of practice for specific competencies (such as complex problem solving), integration of skills across different domains, and, for teaching students how to learn from experience. Interactive stochastic simulations that require students to apply knowledge in order to achieve specific management objectives are especially useful in this respect. Simulations can extend students' learning about theory by providing experience of ill-defined problems in uncertain, dynamic environments, which resemble work in real world business environments. Simulations require students to integrate skills

appropriately across the variety of tasks associated with managing problems over time, such as analyzing complex situations and alternatives, finding information, problem solving, and applying or adapting relevant methods and tools.

Interactive stochastic simulations provide a powerful learning environment for developing the prerequisite self-regulatory skills and attitudes: this in turn would promote the goals of innovation stressed by the AACSB. Simulations can integrate core findings from research on learning to improve the quality and kinds of learning that can be achieved. The interactive nature of the custom designed simulations we used as examples allows participants to receive immediate feedback on the outcomes of their decisions and thus learn from their own level. In addition, students' decisions interact with those of other participants in the simulation so they learn from the actions of others and also experience the behavioral aspects of decision making in an uncertain environment.

The advent of simulation platforms and associated decision cases enable faculty to integrate simulation with other teaching methods across the curriculum. The educational benefits are reinforced when simulations are used on a recurrent basis in courses, extra-curricular events and/or independently accessed by students. We argue that simulations can make an important contribution to the development of the new skills demanded of contemporary management graduates without incurring significant costs in terms of faculty workload or disruption of existing curricula. Students will be better prepared for careers and may also become contributors to the creative process that underlies progress in diverse areas of human endeavor. In business education, as in business, the knowledge and skills to adapt to new situations and to implement innovation effectively can be as important as having innovative and creative ideas and aspirations.

REFERENCES

- AACSB (2018). 2013 *Eligibility Procedures and Accreditation Standards for Business Accreditation*. Retrieved from <https://www.aacsb.edu/~media/aacsb/docs/accreditation/business/standards-and-tables/2018-business-standards.ashx?la=en&hash=B9AF18F3FA0DF19B352B605CBCE17959E32445D9>
- AACSB Task Force (2010). *Business Schools on an Innovation Mission: Report of the AACSB International Task Force on Business Schools and Innovation*. Tampa, Florida: AACSB International. Retrieved from <https://www.aacsb.edu/~media/AACSB/Publications/research-reports/business-schools-on-an-innovation-mission.ashx>
- Alexander, J. C., Heck, C. C. & McElreath, R. B. (2001). A guide to building a university trading

- room. *Financial Services Review*, 10: 209-220.
- Berg, J., Dickaut, J., Hughes, J., McCabe, K. & Rayburn, J. (1995). Capital market experience for financial accounting students. *Contemporary Accounting Research*, 11: 941-958.
- Chi, M.T.H., Feltovich, P.J. & Glaser, R. (1981). Categorization and representation of physics problems by experts and novices. *Cognitive Science*, 5 (2): 121-52.
- Chi, M.T.H., Glaser, R. & Rees, E. (1982). Expertise in problem solving. In R.S. Sternberg (Editor). *Advances in Human Intelligence*: 1-75. Hillsdale, NJ: Erlbaum.
- Clarke, E. (2009). Learning outcomes from business simulation exercises: Challenges for the implementation of learning technologies. *Education and Training*, 51 (5/6): 448-459.
- Coe, T. S. (2007). Using the Bloomberg professional system for finance classes. *Journal of Financial Education*, Spring: 48-62.
- Coe, T., Kilic, O. & Isik, I. (2007). Using an academic trading room to develop a market microstructure course. *Journal of Financial Education*, Winter: 43-52.
- Commission of European Communities (2002). The Lisbon Strategy–Making Change Happen. *Commission Staff Working Paper*. Retrieved from http://www.consilium.europa.eu/uedocs/cms_data/docs/pressdata/en/misc/69861.pdf
- Cook, D.A., Hatala, R., Brydges, R., Zendejas, B., Szostek, J.H., Want, A.T., Erwin, P.J., Hamstra, S.J., 2011. Technology-enhanced simulation for health professions education: a systematic review and meta-analysis. *Journal of the American Medical Association*, 306 (9): 978-988.
- Diamond, R. V. & Smith, H. J. (2011), Threshold concepts: a disciplinary enquiry in quantitative finance. Retrieved from <http://ssrn.com/abstract=1890837>.
- De Jong, T. & Van Joolingen, W. R. (1998). Scientific discovery learning with computer simulations of conceptual domains. *Review of Educational Research*, 68: 179-201.
- Fama, E.F., French, K.R., 2010. Luck versus Skill in the Cross-Section of Mutual Fund Returns. *Journal of Finance*, 65 (5): 1915-1947.
- Gibbs, G. & Simpson, C. (2004). Conditions under which assessment supports students' learning. *Learning and Teaching in Higher Education*, 1: 3-31.
- Harder, B. N. (2010). Use of simulation in teaching and learning in health sciences: a systematic review. *Journal of Nursing Education*, 49(1): 23-28.
- Hattie, J. & Timperley, H. (2007). The power of feedback. *Review of Educational Research* 77: 81-112.
- Hoadley, S., Tickle, L., Wood, L.N. & Kyng, T. (2015). Threshold concepts in finance: conceptualizing the curriculum. *International Journal of Mathematical Education in Science and Technology*, 46 (6): 824-840. DOI: [10.1080/0020739X.2015.1011244](https://doi.org/10.1080/0020739X.2015.1011244)

- Holowczak, R. D. (2005). Incorporating real-time financial data into business curricula. *Journal of Education for Business*, September/October: 3-8.
- Holowczak, R. D. (2007). Real time foreign exchange data modeling in international economics and finance. *Journal of Financial Education*, 33 (Summer): 22-35.
- Issenberg, S.B., McGaghie, W.C., Petrusa, E.R., Lee G.D. & Scalese, R.J., 2005. Features and uses of high-fidelity medical simulations that lead to effective learning: A BEME systematic review. *Med Teach*, 27 (1): 10-28.
- Lombardi, M. M. (2007). Authentic learning for the 21st century: an overview. *Educause Learning Initiative Paper*. Retrieved from <http://net.educause.edu/ir/library/pdf/ELI3009.pdf>
- Lye, A. (2006). Threshold concepts: Reflections on marketing education. In *Proceedings of the 2006 ANZMAC Conference*. Brisbane.
- Mak, K. & McCurdy, T.H. (2019). Simulation-based learning using the RIT market simulator and RIT decision cases. *Journal of Behavioral and Experimental Finance*, 23: 12-22.
- Marton, F. & Säljö, R. (1976). On qualitative differences in learning — 1: outcome and process. *British Journal of Educational Psychology*, 46: 4-11.
- McGahie, W. C., Issenberg, S. B., Petrusa, E. R. & Scalese, R. J. (2010). A critical review of simulation-based medical education research: 2003-2009. *Medical Education*, 44: 50-63.
- Meyer, J. H. F. & Land, R. (2005). Threshold concepts and troublesome knowledge (2): Epistemological considerations and a conceptual framework for teaching and learning. *Higher Education*, 49(3): 373-388.
- Ranchhod, A., Cirai, C. Loukis, E. & Trivedi, R. (2014). Evaluating the educational effectiveness of simulation games: a value generation model. *Information Sciences*, 264: 75-80.
- Rousseau, D. M. & McCarthy, S. (2007). Educating managers from an evidence-based perspective. *Academy of Management Learning & Education*, 6(1): 84-101.
- Salas, E., Wildman, J.L. & Piccolo, R.F., 2009. Using Simulation-Based Training to Enhance Management Education. *Academy of Management Learning & Education* 8 (4): 559-573.
- Siam, J. J. (2005), University trading centres and their role in business education. *Journal of Financial Education*, 31: 11-29.
- Sweller, J. (1994). Cognitive load theory, learning difficulty and instructional design. *Learning and Instruction*, 4: 295-312.
- VanLehn, K. & Chi, M. (2012). Adaptive expertise as acceleration of future learning: a case study. In P.J. Durlach & A.M. Lesgold (Eds.), *Adaptive technologies for training and education*. Cambridge University Press, Cambridge, M.A.: 28-45.

Vermundt, J. D. (2003). The power of learning environments and the quality of student learning. In de Corte, E. Verschaffel, L., Entwistle, N. & van Merrieboer, J. (Eds.), *Powerful learning environments: unraveling basic components and dimensions*: chapter 7. Oxford: Elsevier Science.

Voogt, J., & van den Akker, J. (2002). Computer assisted instruction. *International Encyclopedia of the Social and Behavioral Sciences*: 2473-2477.